**MLA307 Sequence Models**
**Year 2 Semester I**

## Mini Project: 20%

Students will work in groups of maximum 3 members. Each group must identify a real-world Business Intelligence problem involving sequential or time-dependent data and develop a sequence-model-based solution. Your project should demonstrate how sequential data can be collected, analysed, modelled, evaluated, and interpreted to support business decision-making.

**The project should involve:**

1. Identification of a real-world business problem.
2. Collection or selection of an appropriate dataset.
3. Literature review of related work.
4. Selection and implementation of an appropriate sequence model.
5. Evaluation of the model using suitable performance measures.
6. Analysis and interpretation of the results from a business perspective.
7. Recommendations based on the findings.

Students may select one of the following contexts:

### Topic 1: Customer Churn Prediction

A business wants to identify customers who are likely to leave its service. Using historical customer behaviour or interaction data, develop a sequence model to predict customer churn.
Students should:

- Identify sequential patterns in customer behaviour.
- Obtain a suitable customer dataset.
- Develop an RNN, LSTM, or GRU model.
- Train and evaluate the model.
- Identify customers who are at risk of churn.
- Explain how the business could use the results to improve customer retention.

### Topic 2: Time Series Forecasting

A business wants to predict future sales, demand, market trends, or resource utilisation using historical sequential data. Develop an appropriate sequence model to forecast future values.
Students should:

- Identify an appropriate time-series dataset.
- Explain the sequential nature of the data.
- Develop an RNN, LSTM, or GRU model.
- Forecast future values.
- Evaluate the model using appropriate measures such as MAE, MSE, RMSE, or MAPE.
- Explain how the forecast can support business decision-making.

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**Topic 3: Anomaly Detection**
A business wants to identify unusual patterns in sequential data that may indicate fraud, equipment failure, or other abnormal business activities. Develop a sequence-based approach to detect anomalies.

Students may select one of the following contexts:
- Financial transaction fraud
- Machine/sensor failure
- Network activity
- Unusual customer transactions
- Other suitable business-related sequential data

Students should:
- Identify normal and unusual patterns.
- Prepare an appropriate sequential dataset.
- Develop an RNN, LSTM, or GRU-based approach.
- Identify potential anomalies.
- Evaluate the results.
- Explain how anomaly detection could help the business.

**Topic 4: Text Analysis and Sentiment Prediction**
A business receives customer reviews or social-media comments and wants to understand whether customers have positive, negative, or neutral opinions. Develop a sequence-based model to perform sentiment prediction.
Students should:
- Obtain a suitable customer review or text dataset.
- Explain why text can be treated as sequential data.
- Prepare the text data appropriately.
- Develop an RNN, LSTM, or GRU model.
- Predict customer sentiment.
- Evaluate the model.
- Analyse the results and provide business recommendations.

**Topic 5: Recommendation System**
An online business wants to recommend products, services, or content to customers based on their previous behaviour. Using historical customer behaviour or interaction sequences, develop a basic sequence-based recommendation approach.
Students should:
- Identify customer behaviour sequences.
- Analyse previous customer interactions.
- Develop an appropriate sequence-based approach.
- Generate personalised recommendations.
- Evaluate the recommendations using an appropriate approach.
- Explain how recommendations could improve customer engagement or sales.

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**Task 1 – Problem Identification**
Identify a real-world business problem that can be addressed using sequential data.
Explain:

- What is the business problem?
- Why is it important?
- Why is sequential data appropriate?
- What business decision could be supported by the model?

**Task 2 – Literature Review**
Review relevant research papers, articles, or other credible sources related to:

- The selected business problem.
- Sequential data.
- Relevant sequence models.
- Previous approaches.
- Limitations of existing approaches.

**Task 3 – Data Collection and Dataset**
Obtain an appropriate dataset from a credible source.
Explain:

- Source of the dataset.
- Number of records.
- Variables/features.
- Target variable, where applicable.
- Sequential/time-dependent characteristics.
- Data preparation performed.

**Task 4 – Model Selection and Implementation**
Select and justify an appropriate model, such as:

- RNN
- LSTM
- GRU

Implement the selected model using an appropriate programming environment.
Students should explain **why they selected the model** rather than simply using a model because it was demonstrated in class.

**Task 5 – Model Evaluation**
Evaluate the model using appropriate performance measures.
Examples:
**Classification:**

- Accuracy
- Precision
- Recall
- F1-score
- Confusion matrix

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**Regression/forecasting:**

- MAE
- MSE
- RMSE
- MAPE

**Anomaly detection:**

- Precision
- Recall
- F1-score
- Detection rate
- False-positive rate

**Task 6 – Result Analysis**

Interpret the results from a Business Intelligence perspective.
Students should answer questions such as:

- What did the model discover?
- How well did the model perform?
- What sequential patterns were identified?
- What do the results mean for the business?
- How could the organisation use these findings?
- What are the limitations of the model?

**Task 7 – Conclusion and Recommendation**

Provide:

- Main findings.
- Conclusion.
- Business recommendations.
- Possible future improvements.

**Assessment Criteria / Rubric**

| Criteria | Excellent | Good | Satisfactory | Needs Improvement | Marks |
|---|---|---|---|---|---|
| Literature Review | Comprehensive and relevant review with proper citations. | Good review with minor gaps. | Basic review with limited sources. | Inadequate or missing review. | 3 |
| Data Collection & | Appropriate dataset with effective | Suitable dataset with | Dataset selected with | Inappropriate or poorly | 3 |

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|---|---|---|---|---|---|
| Dataset | preprocessing. | adequate preprocessing. | limited preprocessing. | prepared dataset. | |
| Model Selection & Implementation | Appropriate model correctly implemented. | Model implemented with minor errors. | Basic implementation with several errors. | Incorrect or incomplete implementation. | 4 |
| Model Evaluation | Appropriate metrics used with clear evaluation. | Good evaluation with minor gaps. | Basic evaluation with limited interpretation. | Inadequate or missing evaluation. | 4 |
| Result Analysis | Clear insights and strong interpretation. | Good analysis with reasonable interpretation. | Basic analysis with limited insights. | Poor or missing analysis. | 4 |
| Conclusion & Recommendation | Clear conclusions with practical recommendations. | Appropriate conclusions and recommendations. | Basic conclusions with limited recommendations. | Weak or missing conclusion. | 2 |
| Total | | | | | 20 |

**Submission Date:**

October 26, 2026